Independent projectile directions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why horizontal and vertical projectile motion share time but evolve differently.

SUCCESS CRITERIA - I can...

1. Identify horizontal and vertical velocity components.
2. Use the same elapsed time for both axes.
3. State the negligible-air-resistance assumption.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “projectile” and “horizontal” to explain your model or result. Sentence starters are optional.

projectile: an object moving after launch under the stated forces

horizontal: along a level line, usually the x direction

vertical: up and down, usually the y direction

01 Warm-up: bring back an earlier idea

A 3 kg test mass feels 15 N toward a source.
Find g and the force on a 7 kg mass at the same point.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $a_x = 0$; $a_y = -g$; v_x = constant.
Upward positive; g = 10 m/s². Ideal near-Earth motion with negligible air resistance.

Worked reasoning: A ball launches horizontally at 3 m/s and falls for 2 s. Horizontal travel is 3×2=6 m. With g=10 m/s² and no initial vertical velocity, its final vertical velocity is 20 m/s down.

Your turn: Draw separate horizontal and vertical velocity arrows at launch and after 2 s.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

One ball is dropped and another launched horizontally from the same height at the same instant.
Compare fall times and explain.

Sentence starter: Both … . They differ in … because … .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

04 Apply the physics

A ball rolls horizontally off a platform at 3 m/s and is in the air for 2 s.
Label the horizontal and vertical motions.
Find the horizontal travel.
Predict what changes if the horizontal launch speed doubles while the platform height stays the same. Ignore air resistance.

Sentence starter: I predict … . My model supports this because … .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A student says a larger horizontal speed holds the ball up longer.
Explain which force or acceleration would be needed for that claim.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A student says a 4 kg cart under 24 N net force accelerates at 96 m/s².
Identify the error, correct it and name a check you can now do independently.
Draw the forces on the chosen object and label their sources.

Sentence starter: The step I would change is … . The evidence is … .

Add and label force arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Two horizontal launches from the same height have speeds 2 and 6 m/s.
Compare fall times and ranges.
Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.

Sentence starter: Both … . They differ in … because … .

Separate horizontal and vertical motion

Label x and y; state positive directions.

Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two balls launch horizontally from the same platform at 2 and 5 m/s. Their shared fall time is 3 s.
Draw paths and compare ranges.
Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.

Sentence starter: Both … . They differ in … because … .



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3